

Computer Architecture Lab 11 Report

张兆飞

2026-06-16

实验 11 D1 开发板部署神经网络

1. 实验目的

熟悉一个 RISC-V 架构的硬件平台, 并在之上启动一个 Linux 操作系统. 然后在此之上运行 YOLO 算法, 之后需要对卷积层进行优化, 在此过程会使用嵌入的汇编代码进行性能监测, 并对优化的性能进行分析, 在这个过程中我们可以把书中学习到的缓存原理知识应用到实际的工程开发中, 并真切地感受到其发挥的作用. 最后通过 HHB 流程对 RISC-V 的向量扩展的应用有一个初步的认识.

2. 实验步骤

2.1. 将修改过的 Tina Linux 固件烧写到 D1 开发板中.

运行 PhoenixSuit 并执行刷机, 将修改过的 Tina Linux 固件烧写到 D1 开发板中.



图 1: Flash Firmware

2.2. 在 D1 开发板上运行 hello_world 的全流程.

在实验提供的 Ubuntu 虚拟机中, 将本次实验所需的工具添加到环境变量中.

```
loasic@ubuntu: ~  
113 . /usr/share/bash-completion/bash_completion  
114 elif [ -f /etc/bash_completion ]; then  
115 . /etc/bash_completion  
116 fi  
117 fi  
118  
119 # RISC-V GNU Toolchain  
120 export RISC_V_PATH=/opt/riscv-newlib  
121 export RISC_V_LINUX_PATH=/opt/riscv-linux  
122 export PATH=$PATH:${RISC_V_PATH}/bin:${RISC_V_LINUX_PATH}/bin  
123  
124 # QEMU  
125 export RISC_V_QEMU_PATH=/opt/qemu  
126 export PATH=$PATH:${RISC_V_QEMU_PATH}/bin  
127  
128 # D1  
129 export PATH=/home/loasic/experiments/lab11-d1/t-head-linux-toolchain-20210329/bin:$PATH  
130 export PATH=/home/loasic/experiments/lab11-d1/bin:$PATH  
131  
132 alias proxy="source ~/Documents/proxy.sh"  
133  
134  
128,1 Bot
```

图 2: Tool Environment

编译并使用 QEMU 运行 hello_world 程序.

```
loasic@ubuntu: ~/experiments/lab11-d1  
loasic@ubuntu:~/experiments/lab11-d1$ ls  
bin soccfg  
FastestDet t-head-linux-toolchain-20210329  
hello_world tmp  
hello_world.c var  
include XuanTie_CPF_User_Guide.pdf  
lib XuanTie_QEMU_User_Guide.pdf  
libexec xuantie-qemu-x86_64-Ubuntu-18.04-20220402-0353.tar.gz  
Makefile yolo_test_static_new.tar.gz  
share  
loasic@ubuntu:~/experiments/lab11-d1$ which riscv64-unknown-linux-gnu-gcc  
/home/loasic/experiments/lab11-d1/t-head-linux-toolchain-20210329/bin/riscv64-unknown-linux-gnu-gcc  
loasic@ubuntu:~/experiments/lab11-d1$ riscv64-unknown-linux-gnu-gcc -march=rv64i  
mafdcvxtheadc -mabi=lp64dv -mtune=c906 -static -o hello_world hello_world.c  
loasic@ubuntu:~/experiments/lab11-d1$ qemu-riscv64 -cpu c906fdv hello_world  
Hello NeZha  
loasic@ubuntu:~/experiments/lab11-d1$
```

图 3: Compile and Run on QEMU

通过 ADB 工具连接 D1 开发板, 将 hello_world 程序推送到开发板上并运行.

```
jz2024@JZ2024:~$ adb --version
Android Debug Bridge version 1.0.41
Version 34.0.4-debian
Installed as /usr/lib/android-sdk/platform-tools/adb
Running on Linux 6.17.0-35-generic (x86_64)
jz2024@JZ2024:~$ adb devices
List of devices attached
20080411                device

jz2024@JZ2024:~$ adb push Desktop/hello_world /root/hello_world
Desktop/hello_world: 1 file pushed, 0 ...d. 22.8 MB/s (3575576 bytes in 0.149s)
jz2024@JZ2024:~$ adb shell

BusyBox v1.27.2 () built-in shell (ash)

 _--_ _--_ _--_ _--_ _--_ _--_ _--_ _--_ _--_ _--_ _--_
|_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_|
|_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_|
|_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_| |_|
Tina is Based on OpenWrt!

-----
Tina Linux (Neptune, 5C1C9C53)
-----
```

图 4: ADB Connection

[illegible]

图 5: Run on D1

2.3. 撰写带有缓存优化版本的卷积函数, 并通过 QEMU 的测试. (需要写出优化思路, 以及优化的原理.)

根据内存访问的局部性原理优化卷积函数的具体实现. 通过重新排布循环执行顺序, 使得内层循环在遍历整个特征图面时, 始终复用权重切片和输入切片, 有效减少内存的跳跃访问.

通过循环互换, 将当前空间遍历时活跃的计算工作集收敛至 L1 数据缓存大小以内, 解决了顺序流水线因等待内存搬运而产生的数据停顿, 同时避免了 im2col 算法高昂空间开销。

2.4. 在 D1 开发板上进行三个版本的 YOLO 性能测试, 并撰写分析报告. (关于 GEMM 库版本的分析, 同学们需要额外查找资料, 重点放在 `img2col` 算法上. 下面表格中统计的数据除了“执行时间”, 其他的数据可以使用计算量最大的一个层 `conv12` 来代表.)

实验中部分输出如下.

```
Net 12 (convolutional) forwarding...
Function: convolutional_compute
cycles: 47650090376
insts retire: 8108455749
L1 Icache access: 9532620121
L1 Icache access begin: 78072834228602
L1 Icache miss: 3133267
L1 Dcache read access: 1629526759
L1 Dcache read miss: 269999802
L1 Dcache write access: 23152935
L1 Dcache write miss: 949886
Done.
```

图 6: Base Net 12

```
data/dog.jpg: Predicted in 137.669062 seconds.
dog: 57%
car: 52%
truck: 56%
car: 62%
bicycle: 59%
```

图 7: Base Results

```
Net 12 (convolutional) forwarding...
Function: convolutional_compute
cycles: 8050417973
insts retire: 5650425866
L1 Icache access: 5750025831
L1 Icache miss: 591736
L1 Dcache read access: 1606714829
L1 Dcache read miss: 2665211
L1 Dcache write access: 803120296
L1 Dcache write miss: 251929
Done.
```

图 8: GEMM Net 12

```
data/dog.jpg: Predicted in 30.596287 seconds.
dog: 57%
car: 52%
truck: 56%
car: 62%
bicycle: 59%
```

图 9: GEMM Results

```
Net 12 (convolutional) forwarding...
Function: convolutional_compute
cycles: 12197939052
insts retire: 9222033703
L1 Icache access: 9424943796
L1 Icache access begin: 78134079515115
L1 Icache miss: 2015644
L1 Dcache read access: 2883235011
L1 Dcache read miss: 15328544
L1 Dcache write access: 248705424
L1 Dcache write miss: 1481530
Done.
```

图 10: Cache-opt Net 12

```
data/dog.jpg: Predicted in 44.303754 seconds.
dog: 57%
car: 52%
truck: 56%
car: 62%
bicycle: 59%
```

图 11: Cache-opt Results

将上述输出结果整理成表格形式如下.

	Base	GEMM	Cache-opt
cycles	47,650,090,376	8,050,417,973	12,197,939,052
instructions	8,108,455,749	5,650,425,866	9,222,033,703
L1D cache read access	1,629,526,759	1,606,714,829	2,883,235,011
L1D cache read miss	269,999,802	2,665,211	15,328,544
L1D cache read hit ratio	83.43%	99.83%	99.47%
L1D cache write access	23,152,935	803,120,296	248,705,424
L1D cache write miss	949,886	251,929	1,481,530
L1D cache write hit ratio	95.90%	99.97%	99.40%
执行时间 (s)	137.669	30.596	44.304

base 版本使用六层嵌套循环直接计算卷积, 存在对内存数据的跨步访问, 故缓存命中率较低, 运行时间也较长.

gemm 版本使用 im2col 算法将卷积转换为矩阵乘法, 通过调用高效的 GEMM 库实现卷积计算, 大幅提升了性能. 该版本的缓存命中率极高, 因为 im2col 将输入数据重排成连续的内存块, 使得访问模式更友好于缓存系统. 但是 im2col 算法会带来较高的空间开销, 增加了内存的使用量.

cache-opt 版本通过优化循环顺序和数据布局, 使得缓存命中率较 base 版本有显著提升, 有效减少了内存访问的跳跃, 从而降低了执行时间. 该方法引入了多级指针, 带来了额外的内存访问开销, 但由于缓存性能的大幅改善, 整体仍显著优于 base 版本.

2.5. 使用平头哥 HBB 流程编译 mobilenet 生成两个版本的可执行文件 (一个带向量扩展, 一个不带扩展), 并分别在 QEMU 和 D1 开发板运行. 另行选取三张图片重复上述过程, 并将实验结果统计在如下表格中.

在 Ubuntu 虚拟机中通过 docker 启动 HBB 环境.

```

root@dd4950e18611: /
loasic@ubuntu:~$ cd hbb-2.0.20/
loasic@ubuntu:~/hbb-2.0.20$ ls
FastestDet补充代码.rar  hbb.2.0.20.img.tar  start_hhb.sh
loasic@ubuntu:~/hbb-2.0.20$ ./start_hhb.sh
dd4950e18611

  HBB

Welcome to HBB (Heterogeneous Honey Badger) docker

root@dd4950e18611:/# ls
bin  dev  home  lib  media  opt  root  sbin  sys  tools  var
boot  etc  install  lib64  mnt  proc  run  srv  tmp  usr
root@dd4950e18611:/#

```

图 12: HBB Environment

调用 HBB 工具. 将所需 C 语言代码复制到 hbb_out/ 目录下. 修改 C 语言代码并执行编译.

```

root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1# ls
Makefile  cat.jpg  imagenet1000_labels.txt  mobilenetv1.caffemodel  mobilenetv1.prototxt  read_labels.c  read_labels.h  run.sh
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1# hhb -C -cd ./cat.jpg -f ./mobilenetv1.prototxt ./mobilenetv1.caffemodel
-m "103.94 116.98 123.68" -s 0.017 --board c906 --pixel-format BGR
[2026-06-09 15:42:54] (HHB LOG): Start import model.
[2026-06-09 15:42:54] (HHB LOG): Model import completed!
[2026-06-09 15:42:54] (HHB LOG): Start quantization.
[2026-06-09 15:42:54] (HHB LOG): get calibrate dataset from ./cat.jpg
[2026-06-09 15:42:55] (HHB LOG): Start optimization.
[2026-06-09 15:43:00] (HHB LOG): Optimization completed!
[2026-06-09 15:43:00] (HHB LOG): Start conversion to csinn.
[2026-06-09 15:43:00] (HHB LOG): Ignore calibrate dataset in f16/bf16/f32 conversion.
[2026-06-09 15:43:00] (HHB LOG): Conversion completed!
[2026-06-09 15:43:00] (HHB LOG): Start operator fusion.
[2026-06-09 15:43:00] (HHB LOG): Operator fusion completed!
[2026-06-09 15:43:00] (HHB LOG): Start operator split.
[2026-06-09 15:43:00] (HHB LOG): Operator split completed!
[2026-06-09 15:43:00] (HHB LOG): Start layout convert.
[2026-06-09 15:43:00] (HHB LOG): Layout convert completed!
[2026-06-09 15:43:00] (HHB LOG): Quantization completed!
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1# ls
Makefile  hhb_out  mobilenetv1.caffemodel  read_labels.c  run.sh
cat.jpg  imagenet1000_labels.txt  mobilenetv1.prototxt  read_labels.h
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1# cd hhb_out/
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out# ls
data.0.bin  data.0.tensor  hhb.bm  io.c  io.h  main.c  model.c  model.params  process.c  process.h
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out# cp ../read_labels.* ./
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out# cp ../imagenet1000_labels.txt ./
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out# vim main.c
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out# riscv64-unknown-linux-gnu-gcc -O0 -g3 -march=rv64gcv0p7_zfh_xthe
adc -mabi=lp64d -I/home/lib/install_nn2/include -I/home/lib/decode/install/include -o c_runtime_c906 main.c model.c io.c proces
s.c read_labels.c -L/home/lib/install_nn2/lib -L/home/lib/decode/install/lib/rv -ljpeg -lpng -lz -lstdc++ -lshl_c906 -lm -static
-Wl,--gc-sections
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out# ls
c_runtime_c906  data.0.tensor  imagenet1000_labels.txt  io.h  model.c  process.c  read_labels.c
data.0.bin  hhb.bm  io.c  main.c  model.params  process.h  read_labels.h
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out# vim model.c
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out# riscv64-unknown-linux-gnu-gcc -O0 -g3 -march=rv64gcv0p7_zfh_xthe
adc -mabi=lp64d -I/home/lib/install_nn2/include -I/home/lib/decode/install/include -o c_runtime_ref main.c model.c io.c proces
s.c read_labels.c -L/home/lib/install_nn2/lib -L/home/lib/decode/install/lib/rv -ljpeg -lpng -lz -lstdc++ -lshl_ref -lm -static -
Wl,--gc-sections
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out# ls

```

图 13: HBB Compile

使用 QEMU 进行测试.

```

root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out
c_runtime_c906  data.0.bin  hhb.bm  io.c  main.c  model.params  process.h  read_labels.h
c_runtime_ref  data.0.tensor  imagenet1000_labels.txt  io.h  model.c  process.c  read_labels.c
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out# qemu-riscv64 -cpu c906fdv c_runtime_c906 model.params data.0.bin
Run graph execution time: 7451.53955ms, FPS=0.13

=== tensor info ===
shape: 1 3 224 224
data pointer: 0x2212c0

=== tensor info ===
shape: 1 1000 1 1
data pointer: 0x18d3a0
The max_value of output: 0.173218
The min_value of output: 0.000000
The mean_value of output: 0.001009
The std_value of output: 0.000091
===== top5: =====
277(red fox, Vulpes vulpes): 0.173218
282(tiger cat): 0.160156
278(kit fox, Vulpes macrotis): 0.149902
285(Egyptian cat): 0.083435
263(Pembroke, Pembroke Welsh corgi): 0.046417
root@dd4950e18611: /home/example/c906/caffe_mobilenetv1/hhb_out# qemu-riscv64 -cpu c906fdv c_runtime_ref model.params data.0.bin
Run graph execution time: 27758.72070ms, FPS=0.04

=== tensor info ===
shape: 1 3 224 224
data pointer: 0x40010a3010

=== tensor info ===
shape: 1 1000 1 1
data pointer: 0x306760
The max_value of output: 0.163818
The min_value of output: 0.000000
The mean_value of output: 0.000987
The std_value of output: 0.000087
===== top5: =====
277(red fox, Vulpes vulpes): 0.163818
282(tiger cat): 0.159424
278(kit fox, Vulpes macrotis): 0.144043
285(Egyptian cat): 0.088379
281(tabby, tabby cat): 0.049011

```

图 14: QEMU Test

将文件转移至 D1 开发板上, 同样能够正确运行.

```
jz2024@JZ2024:~
root@TinaLinux:~/hbb_out# ./c_runtime_c906 model.params data.0.bin
Run graph execution time: 351.57178ms, FPS=2.84

=== tensor info ===
shape: 1 3 224 224
data pointer: 0x360af760

=== tensor info ===
shape: 1 1000 1 1
data pointer: 0x3601b840
The max_value of output: 0.172119
The min_value of output: 0.000000
The mean_value of output: 0.001009
The std_value of output: 0.000091
===== top5: =====
277(red fox, Vulpes vulpes): 0.172119
282(tiger cat): 0.162354
278(kit fox, Vulpes macrotis): 0.147217
285(Egyptian cat): 0.084229
263(Pembroke, Pembroke Welsh corgi): 0.046143
root@TinaLinux:~/hbb_out# ./c_runtime_ref model.params data.0.bin
Run graph execution time: 40937.23828ms, FPS=0.02

=== tensor info ===
shape: 1 3 224 224
data pointer: 0x3fef87c010

=== tensor info ===
shape: 1 1000 1 1
data pointer: 0x1dd23c00
The max_value of output: 0.163818
The min_value of output: 0.000000
The mean_value of output: 0.000987
The std_value of output: 0.000087
===== top5: =====
277(red fox, Vulpes vulpes): 0.163818
282(tiger cat): 0.159424
278(kit fox, Vulpes macrotis): 0.144043
285(Egyptian cat): 0.088379
281(tabby, tabby cat): 0.049011
root@TinaLinux:~/hbb_out# ls
c_runtime_c906          imagenet1000_labels.txt  model.params
```

图 15: Run on D1

再选取三张图片重复上述过程, 并将实验结果统计在如下表格中.

在不同平台下不同输入的执行时间如下.

	QEMU_ref	QEMU_c906	D1_ref	D1_c906
图片 1	25284.25	7387.89	41164.05	358.48
图片 2	25354.57	7060.39	41194.71	607.63
图片 3	25198.65	7025.86	41194.57	358.56

在不同平台下不同输入的分类结果如下.

	QEMU_ref	QEMU_c906	D1_ref	D1_c906
图片 1	354 Arabian camel	354 Arabian camel	354 Arabian camel	354 Arabian camel
图片 2	207 Golden retriever	207 Golden retriever	207 Golden retriever	207 Golden retriever
图片 3	404 Airliner	404 Airliner	404 Airliner	404 Airliner

2.6. 完成手势识别网络的后处理函数编程, 使用 D1 开发板部署手势识别网络. 要求提供对 test_case 中测试图片的输出图片.

调用 HHB 进行网络编译.


```
root@dd4950e18611:/home/example/c906/FastestDet# ls
FastestDet.onnx camera_utils.c camera_utils.h post_process.c post_process.h
stb_image.h stb_image_write.h test_case
root@dd4950e18611:/home/example/c906/FastestDet# hhb -C --model-file ./FastestDe
t.onnx --input-name input --output-name output --input-shape "1 3 320 320" --dat
a-scale 0.003922 --postprocess save --quantization-scheme "float32" --board c906
[2026-06-08 13:55:22] (HHB LOG): Start import model.
[2026-06-08 13:55:25] (HHB LOG): Model import completed!
[2026-06-08 13:55:25] (HHB LOG): Start quantization.
[2026-06-08 13:55:25] (HHB LOG): Start optimization.
[2026-06-08 13:55:25] (HHB LOG): Optimization completed!
[2026-06-08 13:55:25] (HHB LOG): Start conversion to csinn.
[2026-06-08 13:55:26] (HHB LOG): Conversion completed!
[2026-06-08 13:55:26] (HHB LOG): Start operator fusion.
[2026-06-08 13:55:26] (HHB LOG): Operator fusion completed!
[2026-06-08 13:55:26] (HHB LOG): Start operator split.
[2026-06-08 13:55:26] (HHB LOG): Operator split completed!
[2026-06-08 13:55:26] (HHB LOG): Start layout convert.
[2026-06-08 13:55:26] (HHB LOG): Layout convert completed!
[2026-06-08 13:55:27] (HHB LOG): Quantization completed!
root@dd4950e18611:/home/example/c906/FastestDet# ls
FastestDet.onnx camera_utils.h post_process.c stb_image.h test_case
camera_utils.c hhb_out post_process.h stb_image_write.h
root@dd4950e18611:/home/example/c906/FastestDet#
```

图 16: HHB Compile

将所需 C 语言代码复制到 hhb_out/ 目录下. 修改 C 语言代码, 并补全非极大值抑制算法的实现.

```
root@dd4950e18611:/home/example/c906/FastestDet# cp post* stb* hhb_out/
root@dd4950e18611:/home/example/c906/FastestDet# ll
total 1068
drwxr-xr-x 5 root root 4096 Jun 8 14:03 ./
drwxr-xr-x 1 root root 4096 Dec 7 2023 ../
drwxr-xr-x 2 root root 4096 Dec 7 2023 .pk1_memoize_py3/
-rwxrwxrwx- 1 1000 1000 685811 Dec 8 2023 FastestDet.onnx*
-rwxr--r-- 1 root root 3048 Dec 11 2023 camera_utils.c*
-rwxr--r-- 1 root root 672 Dec 11 2023 camera_utils.h*
drwxr-xr-x 2 root root 4096 Jun 8 14:29 hhb_out/
-rwxrwxrwx- 1 1000 1000 13479 Dec 12 2023 post_process.c*
-rwxrwxrwx- 1 1000 1000 2798 Dec 11 2023 post_process.h*
-rwxrwxrwx- 1 1000 1000 278740 Dec 10 2023 stb_image.h*
-rwxrwxrwx- 1 1000 1000 67387 Dec 10 2023 stb_image_write.h*
drwxr-xr-x 2 root root 4096 Dec 11 2023 test_case/
root@dd4950e18611:/home/example/c906/FastestDet# cd hhb_out/
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# ll
total 1908
drwxr-xr-x 2 root root 4096 Jun 8 14:29 ./
drwxr-xr-x 5 root root 4096 Jun 8 14:03 ../
-rw-r--r-- 1 root root 636400 Jun 8 13:55 hhb.bm
-rw-r--r-- 1 root root 4946 Jun 8 13:55 io.c
-rw-r--r-- 1 root root 1539 Jun 8 13:55 io.h
-rw-r--r-- 1 root root 6638 Jun 8 13:55 main.c
-rw-r--r-- 1 root root 257563 Jun 8 13:55 model.c
-rw-r--r-- 1 root root 628208 Jun 8 13:55 model.params
-rwxr--r-- 1 root root 13479 Jun 8 14:29 post_process.c*
-rwxr--r-- 1 root root 2798 Jun 8 14:29 post_process.h*
-rw-r--r-- 1 root root 20086 Jun 8 13:55 process.c
-rw-r--r-- 1 root root 2040 Jun 8 13:55 process.h
-rwxr--r-- 1 root root 278740 Jun 8 14:29 stb_image.h*
-rwxr--r-- 1 root root 67387 Jun 8 14:29 stb_image_write.h*
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# vim main.c
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# vim main.c
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# vim post_process.c
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# touch new_post_process.c
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# vim new_post_process.c
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# mv post_process.c post_process.c.backup
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# ll
total 1928
drwxr-xr-x 2 root root 4096 Jun 8 14:34 ./
drwxr-xr-x 5 root root 4096 Jun 8 14:03 ../
```

图 17: Modify Source Code

执行编译. 将上级目录下的 test_case/ 复制到 hhb_out/ 目录下, 随后使用 QEMU 进行测试.


```

root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# mv new_post_process.c post_process.c
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# riscv64-unknown-linux-gnu-gcc -O0 -g3 -march=rv64gcv0p7_zfh_xtheadc -ma
bi-lp64d -I/home/lib/install_nn2/include -I/home/lib/decode/install/include -o c_runtime_c906 main.c model.c io.c process.c pos
t_process.c -L/home/lib/install_nn2/lib -L/home/lib/decode/install/lib/rv -ljpeg -lpng -lz -lstdc++ -lshl_c906 -lm -static -Wl,-
-gc-sections
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# echo $?
0
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# ll ..
total 1068
drwxr-xr-x 5 root root 4096 Jun 8 14:03 ./
drwxr-xr-x 1 root root 4096 Dec 7 2023 ../
drwxr-xr-x 2 root root 4096 Dec 7 2023 .pk1_memoize_py3/
-rwxr--r-- 1 1000 1000 685811 Dec 8 2023 FastestDet.onnx*
-rwxr--r-- 1 root root 3048 Dec 11 2023 camera_utils.c*
-rwxr--r-- 1 root root 672 Dec 11 2023 camera_utils.h*
drwxr-xr-x 2 root root 4096 Jun 8 14:35 hhb_out/
-rwxr--r-- 1 1000 1000 13479 Dec 12 2023 post_process.c*
-rwxr--r-- 1 1000 1000 2798 Dec 11 2023 post_process.h*
-rwxr--r-- 1 1000 1000 278740 Dec 10 2023 stb_image.h*
-rwxr--r-- 1 1000 1000 67387 Dec 10 2023 stb_image_write.h*
drwxr-xr-x 2 root root 4096 Dec 11 2023 test_case/
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# cp -r ../test_case/ ./
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# qemu-riscv64 -cpu c906fdv c_runtime_c906 model.params test_case/test_ca
se.txt
Run graph execution time: 807.67212ms, FPS=1.24

=== tensor info ===
shape: 1 3 320 320
data pointer: 0x437d70

=== tensor info ===
shape: 1 11 20 20
data pointer: 0xf9af0
GenerateOutputPicture
output_bboxes[0]: 0.399967
output_bboxes[1]: 0.476099
output_bboxes[2]: 0.103089
output_bboxes[3]: 0.137135
output_bboxes[4]: 0.959916
output_bboxes[5]: 0.000000
-----
Is Gesture 1

```

图 18: Compile and Test

所有图片中的手势均被正确识别。

```

data pointer: 0x263360

=== tensor info ===
shape: 1 11 20 20
data pointer: 0x2056a0
GenerateOutputPicture
output_bboxes[0]: 0.378068
output_bboxes[1]: 0.568738
output_bboxes[2]: 0.121473
output_bboxes[3]: 0.204555
output_bboxes[4]: 0.898585
output_bboxes[5]: 4.000000
-----
Is Gesture 1
Run graph execution time: 924.14923ms, FPS=1.08

=== tensor info ===
shape: 1 3 320 320
data pointer: 0x263360

=== tensor info ===
shape: 1 11 20 20
data pointer: 0x2056a0
GenerateOutputPicture
output_bboxes[0]: 0.541847
output_bboxes[1]: 0.576802
output_bboxes[2]: 0.162334
output_bboxes[3]: 0.132248
output_bboxes[4]: 0.964846
output_bboxes[5]: 5.000000
-----
Is Gesture 1
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# ls
0.jpg_output0_0_0.959916.jpg 3.jpg_output0_1_11_20_20.txt c_runtime_c906 model.params stb_image.h
0.jpg_output0_1_11_20_20.txt 3.jpg_output0_3_0.808383.jpg hhb.bm post_process.c stb_image_write.h
1.jpg_output0_1_0.889453.jpg 4.jpg_output0_1_11_20_20.txt io.c post_process.c.backup test_case
1.jpg_output0_1_11_20_20.txt 4.jpg_output0_4_0.898585.jpg io.h post_process.h
2.jpg_output0_1_11_20_20.txt 5.jpg_output0_1_11_20_20.txt main.c process.c
2.jpg_output0_2_0.888378.jpg 5.jpg_output0_5_0.964846.jpg model.c process.h
root@dd4950e18611:/home/example/c906/FastestDet/hhb_out# exit

```

图 19: Test Results

使用简单的 BitMap, 将识别结果可视化在输出图片上. 最终输出的图片如下.

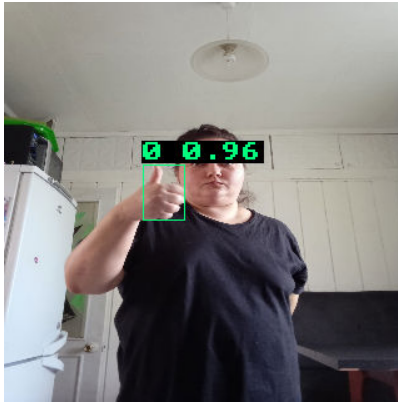


图 20: Thumbs Up



图 21: Gesture 1

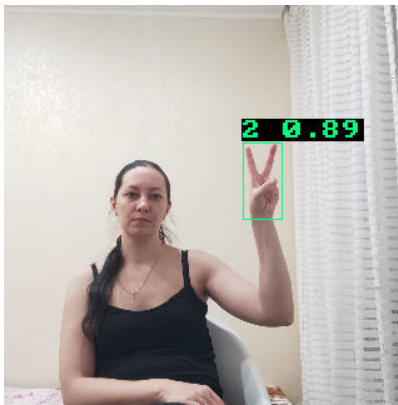


图 22: Gesture 2



图 23: Gesture 3



图 24: Gesture 4

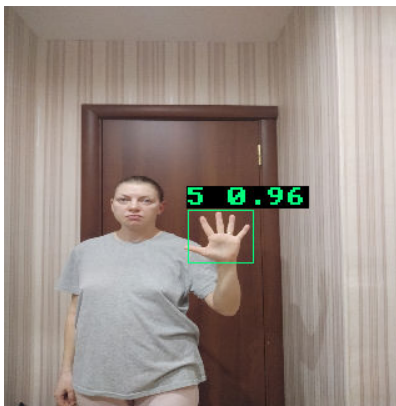


图 25: Gesture 5

将文件转移至 D1 开发板上, 同样能够正确识别手势并输出结果.

```

root@TinaLinux:~/hbb_out_6_bitmap# ls
c_runtime_c906      model.c              process.c
hbb.bm             model.params         process.h
io.c               post_process.c       stb_image.h
io.h               post_process.c.backup stb_image_write.h
main.c             post_process.h        test_case
root@TinaLinux:~/hbb_out_6_bitmap# ./c_runtime_c906 model.params test_case/test_
case.txt
Run graph execution time: 336.53806ms, FPS=2.97

=== tensor info ===
shape: 1 3 320 320
data pointer: 0xe46f220

=== tensor info ===
shape: 1 11 20 20
data pointer: 0xe230fa0
GenerateOutputPicture
output_bboxes[0]: 0.399967
output_bboxes[1]: 0.476099
output_bboxes[2]: 0.103089
output_bboxes[3]: 0.137135
output_bboxes[4]: 0.959916
output_bboxes[5]: 0.000000
-----
Is Gesture      1
-----
Run graph execution time: 327.59644ms, FPS=3.05

=== tensor info ===
shape: 1 3 320 320
data pointer: 0xe29a810

=== tensor info ===
shape: 1 11 20 20
data pointer: 0xe2355b0
GenerateOutputPicture
output_bboxes[0]: 0.143534
output_bboxes[1]: 0.309319
output_bboxes[2]: 0.157776
output_bboxes[3]: 0.300725
output_bboxes[4]: 0.889453

```

图 26: Run on D1

2.7. D1 开发板调用摄像头组件, 完成实时手势识别, 对六种识别手势分别进行测试.

修改 main.c, 并且将 camera_utils.c 和 camera_utils.h 搬运至 hbb_out/ 目录下. 接着调用平头哥工具链完成编译. 在 D1 开发板上运行运行该程序, 摄像头进行图像捕捉. 在开发板上得到运行结果, 能够成功检测出手势.

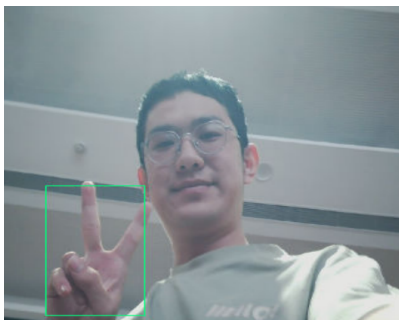


图 27: Real-time Detection

3. 实验分析与总结

本次实验通过在平头哥 C906 处理器上运行不同版本的 YOLO 卷积层以及部署 MobileNet 和 FastestDet 网络, 深入分析了软硬件协同优化对推理效率的巨大提升效果. 在 YOLO 卷积层性能测试中, 通过数据重排的 Cache-opt 版本利用内存访问的局部性原理, 有效降低了 L1D 缓存数据读取的缺失率, 显著缩短了整体执行时间. 而借助 im2col 算法转换矩阵运算的 GEMM 版本则在空间开销与极致计算效率之间做出了权衡, 取得了最短的运行耗时. 此外, 在 MobileNet 的模型部署对比测试中, 链接了 RISC-V 向量扩展指令集算子库的 C906 版本相较于无硬件加速的 ref 版本, 在真实的 D1 开发板上实现了极大的推理速度提升, 充分证明了硬件加速的在特定场景下的显著优势.

本次实验覆盖了从底层嵌入式系统到上层神经网络部署的全流程. 实验首先基于 Tina Linux 完成了 D1 开发板的基础环境搭建, 利用平头哥交叉编译工具链与 QEMU 实现了 C 语言裸机程序的编译与功能验证, 并将可执行文件通过 ADB 工具传输至开发板. 随后, 实验从软件仿真跨越到了真实环境部署, 不仅手动验证了缓存优化在 YOLO 网络计算中的工程价值, 还借助了 HHB 编译框架将前端模型转化为能在目标机器上运行的 C 代码. 最终, 通过自主编写非极大值抑制后处理代码并成功调用摄像头, 系统在 D1 平台上实现了实时手势识别的完整功能链路.

4. 实验收获与建议

本次综合性较强的硬件与 AI 交叉实验带来了丰富的理论与实践经验, 将体系结构中抽象的缓存命中与缺失原理真实地应用到了卷积计算的代码优化中, 深刻体会到了数据局部性原理对系统整体吞吐率的影响. 同时, 通过对比 ref 与开启向量扩展的 C906 数据模型, 直观地领略到了 RISC-V 架构通过硬件层面的向量指令和算数增强扩展, 在应对矩阵运算与图像处理时的强大底层硬件优势. 此外, 熟练掌握设备控制, 交叉编译工作流, HBB 算子转化流程, 以及处理复杂的多维数据内存变换, 极大拓宽了嵌入式 Linux 开发和软硬件协同开发的实际应用能力.

本次实验内容较少且部分步骤需要使用开发板, 在十几人的小组合作中难以保证每个同学都能亲自进行实验操作. 建议在硬件资源允许的情况下减少单组人数, 提升实验的参与度. 此外, 实验最后一步依赖特定摄像头组件, 然而由于没有足够的摄像头供每组使用, 实验进度严重受限. 建议增加实验相关设备的数量, 或者提供更多的替代方案, 以确保每个同学都能顺利进行实验.